LITERATURE REVIEW

Design of a Telemonitoring System for Patients with Chronic Diseases (Hypertension & Diabetes) Using Consultation Data, Home Measurements, and Mobile Sensors

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Nineteen peer-reviewed sources, grouped into six thematic clusters aligned with the Chapter 2 structure. APA format; every citation checked against PubMed, PMC, Crossref and publisher records, and every DOI resolves.

CLUSTER 1 — EPIDEMIOLOGICAL BURDEN: HYPERTENSION AND DIABETES IN AFRICA

#	Source & Citation	Problem Addressed	Research Contribution	Aim & Objectives	Limitations	Findings & Conclusions	Methods	My Abstract / Relevance
1	**Hinneh et al. (2023) — PLOS Global Public Health** "Regional Prevalence of Hypertension Among People Diagnosed with Diabetes in Africa: A Systematic Review and Meta-Analysis" Hinneh, T., et al. (2023). Regional prevalence of hypertension among people diagnosed with diabetes in Africa, a systematic review and meta-analysis. *PLOS Global Public Health, 3*(12), e0001931. https://doi.org/10.1371/journal.pgph.0001931	The regional burden of hypertension-diabetes comorbidity across Africa was fragmented across many small, heterogeneous single-country studies, with no pooled regional estimate to guide health system planning.	The first systematic quantification of hypertension prevalence specifically within diabetic populations by African sub-region, pooling 41 studies covering over 116,000 patients.	Estimate overall and region-specific pooled prevalence of hypertension among people with diabetes in Africa; identify sub-regional disparities and factors associated with higher comorbidity risk.	High heterogeneity across included studies; mostly cross-sectional designs, precluding causal inference; inconsistent diagnostic criteria. The Central African estimate carries a wide confidence interval (95% CI 53.0–91.4%).	Pooled hypertension prevalence among people with diabetes in Africa was 58.1%, with **Central Africa highest at 77.6%** — above Southern (69.1%), North (63.4%), East (53.0%) and West Africa (51.5%). Older age, overweight/obesity, longer diabetes duration, urban residence and male sex were associated with higher risk.	Systematic review and random-effects meta-analysis following PRISMA; EMBASE, PubMed and HINARI searched to March 2023; Mixed Methods Appraisal Tool; I^2 statistics and funnel plots.	The foundational epidemiological justification for this thesis: the highest documented hypertension-diabetes comorbidity burden on the continent falls in exactly this study's deployment region. This is the strongest available evidence that a combined dual-condition telemonitoring system, rather than a single-condition tool, is the right clinical target. Report the confidence interval alongside the 77.6% figure. → *Chapter 1 §1.1 (Background/Motivation); Chapter 2 §2.1–2.2 (Epidemiological Burden)*
2	**Gafane-Matemane et al. (2024) — Journal of Human Hypertension** "Hypertension in Sub-Saharan Africa: The Current Profile, Recent Advances, Gaps, and Priorities" Gafane-Matemane, L. F., Craig, A., Kruger, R., Alaofin, O. S., Ware, L. J., Jones, E. S. W., & Kengne, A. P. (2024). Hypertension in sub-Saharan Africa: The current profile, recent advances, gaps, and priorities. *Journal of Human Hypertension, 39*(2), 95–110. https://doi.org/10.1038/s41371-024-00913-6	High and increasing prevalence of hypertension in sub-Saharan Africa with persistently poor detection, treatment and control rates.	Narrative synthesis of the SSA hypertension burden for 2017–2023, community-based management strategies, prominent risk factors, and underlying mechanisms of hypertension development in the region.	Summarise the regional burden; review community-based management strategies; outline risk factors and mechanisms; identify gaps and research priorities.	Lack of nationally representative studies; heterogeneity of sampling and data collection across countries; some countries have no data for the reporting period. Narrative rather than pooled synthesis.	Most SSA countries reported hypertension prevalence during 2017–2023, confirming high burden and poor control. Task-shifting to community health workers improved linkage to care and medication adherence, though BP-lowering effects over time were inconsistent.	Narrative review; literature search January 2017 – June 2023 across MEDLINE, PubMed, CINAHL, AJOL and grey literature.	Establishes that *detection and control* — not prevalence alone — are where SSA health systems fail, which is precisely the gap a continuity mechanism addresses. The inconsistent results of task-shifting strengthen the case for technology-supported follow-up rather than personnel-only interventions. → *Chapter 1 §1.1 (Background); Chapter 2 §2.2 (Regional Burden)*
3	**World Health Organization (2023)** "Global Report on Hypertension: The Race Against a Silent Killer" World Health Organization. (2023). *Global report on hypertension: The race against a silent killer.* Geneva: World Health Organization. https://www.who.int/publications/i/item/9789240081062	Hypertension remains a leading cause of preventable cardiovascular death globally, and progress against the global NCD target for uncontrolled hypertension has been insufficient.	WHO's first dedicated global report on hypertension: consolidated global epidemiology, country-level surveillance data, and policy recommendations for detection, treatment and control.	Map the global hypertension burden; set policy targets; recommend multi-level interventions for health systems worldwide.	Population-level and policy-oriented; does not address patient-level monitoring technology, implementation detail, or low-resource deployment specifically. Institutional report rather than peer-reviewed literature.	Approximately one billion people worldwide have hypertension, with around two-thirds living in low- and middle-income countries. Digital health and community-based strategies are endorsed among the recommended responses.	Global surveillance data; country-level epidemiological analysis; policy synthesis.	Provides the global framing and institutional endorsement of digital and community-level approaches, situating this project inside a recognised strategy rather than presenting it as a novel gamble. Cite as institutional/grey literature, distinct from the peer-reviewed sources. → *Chapter 1 §1.1 (Background); Chapter 2 §2.1 (Global Burden)*

CLUSTER 2 — mHEALTH SELF-MANAGEMENT AND CLINICAL OUTCOME VALIDATION

#	Source & Citation	Problem Addressed	Research Contribution	Aim & Objectives	Limitations	Findings & Conclusions	Methods	My Abstract / Relevance
4	**Mihevc et al. (2025) — JMIR mHealth and uHealth** "Impact of 12-Month mHealth Home Telemonitoring on Clinical Outcomes in Older Individuals With Hypertension and Type 2 Diabetes: Multicenter Randomized Controlled Trial" Mihevc, M., Mori Lukančič, M., Zavrnik, Č., Virtič Potočnik, T., Ružić Gorenjec, N., Petek Šter, M., Klemenc-Ketiš, Z., & Poplas Susič, A. (2025). Impact of 12-month mHealth home telemonitoring on clinical outcomes in older individuals with hypertension and type 2 diabetes: Multicenter randomized controlled trial. *JMIR mHealth and uHealth, 13*, e59733. https://doi.org/10.2196/59733	Limited evidence on the clinical effectiveness of mHealth home telemonitoring for older people with both arterial hypertension and type 2 diabetes in primary care settings.	A 12-month multicenter open-label randomised controlled trial in primary care comparing mHealth home telemonitoring against standard care in patients aged 65 and over with both conditions.	Evaluate the impact of 12-month telemonitoring on clinical outcomes in older individuals with hypertension and type 2 diabetes compared to standard care in a primary care setting.	Conducted in Slovenia — a high-income European setting — so applicability to LMICs requires adaptation; older-adult population limits generalisability to younger patients; open-label design.	Telemonitoring **effectively improves hypertension and type 2 diabetes control** in older people, **but has no impact** on other cardiovascular risk factors or diabetes-related quality of life. Combining educational and behavioural components with telemonitoring is recommended for further study.	Multicenter, open-label RCT with 1:1 randomisation by an independent third party; 12-month follow-up; blood pressure measured twice weekly, morning and evening.	The most directly relevant clinical trial available — same dual-condition population, same intervention modality. Report the null result alongside the positive one: telemonitoring improved disease control but not broader risk factors or quality of life. That nuance is more credible than an unqualified endorsement and supports pairing monitoring with patient education. → *Chapter 2 §2.3 (Clinical Evidence); Chapter 6 (Discussion & Expected Impact)*
5	**Mao et al. (2020) — BMJ Open Diabetes Research & Care** "Impact and Efficacy of Mobile Health Intervention in the Management of Diabetes and Hypertension" Mao, Y., Lin, W., Wen, J., & Chen, G. (2020). Impact and efficacy of mobile health intervention in the management of diabetes and hypertension: A systematic review and meta-analysis. *BMJ Open Diabetes Research & Care, 8*(1), e001225. https://doi.org/10.1136/bmjdrc-2020-001225	Insufficient synthesis of evidence on whether mHealth interventions effectively improve clinical outcomes across patients managing both diabetes and hypertension simultaneously.	Combined systematic review and meta-analysis quantifying the clinical impact of mHealth on HbA1c, fasting blood glucose and blood pressure in dual-condition patients.	Quantify the pooled effect of mHealth on blood glucose and blood pressure control; assess moderating factors including app features and follow-up duration.	Publication bias likely; heterogeneity in intervention types; most trials short-term; limited representation of low-resource settings.	mHealth interventions significantly reduced HbA1c and systolic blood pressure in patients with diabetes and/or hypertension. Combined condition management via mobile platforms is clinically effective.	Systematic review and meta-analysis; random-effects model; subgroup analyses by condition, intervention type and follow-up period.	Provides the pooled quantitative evidence that mobile health interventions meaningfully improve both glycaemic and blood pressure outcomes simultaneously — the core clinical goal of the proposed dual-monitoring design. → *Chapter 2 §2.3 and the abstract; supports the dual-condition monitoring rationale*
6	**Bandeira et al. (2024) — Telemedicine and e-Health** "Effect of m-Health-Based Interventions on Blood Pressure: An Updated Systematic Review with Meta-Analysis" Bandeira, A. C. N., de Melo, P. U. G., Johann, E. B., Ritti-Dias, R. M., Rech, C. R., & Gerage, A. M. (2024). Effect of m-health-based interventions on blood pressure: An updated systematic review with meta-analysis. *Telemedicine and e-Health, 30*(4), 987–1001. https://doi.org/10.1089/tmj.2023.0545	Prior meta-analyses on mHealth and blood pressure predate recent randomised trial evidence; an updated synthesis was needed.	Updated meta-analysis of mHealth interventions on blood pressure, incorporating recent RCTs with subgroup analyses by intervention modality, population and follow-up duration.	Quantify the updated pooled effect of mHealth on blood pressure; evaluate moderation by intervention features; inform clinical guideline updates.	Heterogeneity in intervention types; short follow-up in several trials; limited data from LMICs and low-connectivity settings.	mHealth interventions produce statistically and clinically significant blood pressure reductions across diverse populations and settings. **Telemonitoring with clinician feedback is the most effective modality.**	Systematic review; random-effects meta-analysis; subgroup analysis; risk-of-bias assessment.	The most current quantitative evidence that mHealth telemonitoring reduces blood pressure. Its specific finding that clinician feedback is the most effective modality is direct support for the clinician-in-the-loop architecture rather than a patient-only self-tracking application. → *Chapter 2 §2.3; Chapter 6 §6.6 (Clinical Alignment Discussion)*
7	**Alessa et al. (2019) — JMIR mHealth and uHealth** "Smartphone Apps to Support Self-Management of Hypertension: Review and Content Analysis" Alessa, T., Hawley, M. S., Hock, E. S., & de Witte, L. (2019). Smartphone apps to support self-management of hypertension: Review and content analysis. *JMIR mHealth and uHealth, 7*(5), e13645. https://doi.org/10.2196/13645	The rapid proliferation of commercial hypertension apps left patients and clinicians with no basis for judging which apps contain evidence-based content or adequate privacy safeguards.	Systematic identification and content analysis of commercially available hypertension self-management apps, coded against recommended self-management components and quality/security criteria.	Identify hypertension apps available to the public and analyse which self-management functions, evidence base and privacy provisions they actually contain.	App marketplaces change continuously, so the sample dates quickly; content analysis assesses features present rather than clinical effectiveness; English-language apps predominate.	Most available apps support only a narrow set of self-management functions; few are grounded in clinical evidence or address privacy adequately. Feature completeness and trustworthiness vary widely across the market.	Systematic app-store search; structured content analysis against self-management and quality criteria.	Documents what app-based hypertension self-management typically does and does not include — useful for positioning the feature set of the patient-facing application, and for arguing that a clinician-linked, evidence-grounded design is not the market norm. → *Chapter 2 §2.3 (Mobile Health); Chapter 3 (UX Design Rationale)*
8	**Zhou et al. (2024) — Journal of Medical Internet Research** "Behavior Change Techniques Used in Self-Management Interventions Based on mHealth Apps for Adults With Hypertension: Systematic Review and Meta-Analysis of Randomized Controlled Trials" Zhou, Y., Li, S.-J., Huang, R.-Q., Ma, H.-M., Wang, A.-Q., Tang, X.-Y., Pei, R.-Y., & Piao, M.-H. (2024). Behavior change techniques used in self-management interventions based on mHealth apps for adults with hypertension: Systematic review and meta-analysis of randomized controlled trials. *Journal of Medical Internet Research, 26*, e54978. https://doi.org/10.2196/54978	Evidence on mHealth app self-management for hypertension needed updating, and the active ingredients that actually drive behaviour change within these apps remained unclear.	Meta-analysis linking specific behaviour change techniques embedded in mHealth hypertension apps to measured blood pressure outcomes.	Evaluate the effect of mHealth app self-management interventions on blood pressure; investigate which behaviour change techniques are used and how they relate to effect size.	Behaviour change technique coding is inherently interpretive; short follow-up in many RCTs; limited LMIC representation; publication bias.	mHealth app self-management interventions are effective for lowering blood pressure in hypertensive patients, and effect size is influenced by intervention design and the behaviour change techniques employed.	Systematic review and meta-analysis of RCTs across six databases (January 2009 – October 2023); Cochrane Risk of Bias v2; behaviour change technique taxonomy coding; PROSPERO CRD42023483746.	The evidence base for the feature set rather than the model: self-monitoring, feedback and goal-setting are the components with measurable blood pressure effect, which justifies the alert engine and trend visualisation in the patient application. → *Chapter 2 §2.3; Chapter 3 (Feature Justification); Chapter 5 (App Design)*

CLUSTER 3 — IoT AND SENSOR TECHNOLOGY FOR CHRONIC DISEASE MONITORING

#	Source & Citation	Problem Addressed	Research Contribution	Aim & Objectives	Limitations	Findings & Conclusions	Methods	My Abstract / Relevance
9	**Tai et al. (2024) — Transactions on Emerging Telecommunications Technologies** "Internet of Things-Enabled Intelligent Systems for Remote Chronic Disease Monitoring" Tai, C., Goyal, S. B., Bedi, P., & Amannah, C. (2024). Internet of things-enabled intelligent systems for remote chronic disease monitoring. *Transactions on Emerging Telecommunications Technologies, 35*(4), e4919. https://doi.org/10.1002/ett.4919	Conventional machine learning struggles with IoT health data because it is noisy, heterogeneous and high-dimensional; no standardised architecture existed for remote chronic disease monitoring.	Proposes a human-centric deep-learning technique for remote chronic disease monitoring in an IoT context, together with a standardised three-phase architecture: data acquisition → data processing → data visualisation.	Develop a technique able to handle complex, heterogeneous IoT data; produce accurate and interpretable predictions and recommendations; define a replicable architecture.	Evaluated in controlled conditions; real-world deployment under the connectivity constraints of low-resource settings is not addressed.	The three-phase architecture supports accurate prediction and actionable clinical recommendations for remotely monitored patients, while remaining interpretable.	Deep learning applied to IoT health data; standardised architecture design; performance evaluation.	Directly informs this system's architecture: acquisition (mobile app) → processing (cloud backend and on-device model) → visualisation (clinician dashboard). Note that Tai et al. process centrally with deep learning whereas this thesis runs a classical ensemble on-device — a deliberate divergence justified by the connectivity constraint their own limitations acknowledge. *→ Chapter 2 §2.4 (IoT Architecture); Chapter 3 §3.5 (System Architecture)*
10	**Liu & Wang (2025) — Frontiers in Public Health** "Advanced Applications in Chronic Disease Monitoring Using IoT Mobile Sensing Device Data, Machine Learning Algorithms and Frame Theory: A Systematic Review" Liu, Y., & Wang, B. (2025). Advanced applications in chronic disease monitoring using IoT mobile sensing device data, machine learning algorithms and frame theory: A systematic review. *Frontiers in Public Health*, *13*, 1510456. https://doi.org/10.3389/fpubh.2025.1510456	Fragmented evidence on how IoT sensing devices and machine learning algorithms are applied across different chronic diseases, without a unified architectural or theoretical framework.	Systematic review integrating IoT mobile sensing, machine learning and tight-frame methods for chronic disease monitoring, covering cardiovascular disease, cancer, chronic respiratory disease and diabetes.	Survey IoT and ML applications for chronic disease monitoring; evaluate their role in risk assessment and personalised management; synthesise cross-condition best practice.	Controlled settings dominate the reviewed studies; real-world deployment in LMICs is not well represented; interoperability is not fully addressed.	IoT devices combined with ANN, SVM, Random Forest and deep learning support risk assessment and personalised management across chronic conditions; real-time monitoring architectures are the dominant pattern.	Systematic database search with defined inclusion/exclusion criteria; quality assessment; machine learning algorithm comparison.	The most current cross-condition synthesis of IoT and ML for chronic disease monitoring. Its explicit inclusion of Random Forest among the effective algorithm families supports the classical-ensemble choice made here, and its LMIC deployment gap reinforces the Central African contribution. *→ Chapter 2 §2.4–2.5; Chapter 3 §3.5 (System Architecture)*
11	**Alhaddad et al. (2022) — Frontiers in Bioengineering and Biotechnology** "Sense and Learn: Recent Advances in Wearable Sensing and Machine Learning for Blood Glucose Monitoring and Trend-Detection" Alhaddad, A. Y., Aly, H., Gad, H., Al-Ali, A., Sadasivuni, K. K., Cabibihan, J.-J., & Malik, R. A. (2022). Sense and learn: Recent advances in wearable sensing and machine learning for blood glucose monitoring and trend-detection. *Frontiers in Bioengineering and Biotechnology, 10*, 876672. https://doi.org/10.3389/fbioe.2022.876672	Growing demand for non-invasive, continuous blood glucose monitoring is not met by existing clinical tools; the wearable sensor literature of the past five years lacked synthesis.	Comprehensive review of emerging non-invasive sensing modalities (ECG, electromagnetic, bioimpedance, photoplethysmography) and the machine learning models applied to them for glucose estimation and trend detection.	Review sensor technologies and ML methods for non-invasive blood glucose monitoring and hypoglycaemia detection; identify challenges and recommend future directions.	Many reviewed studies have small samples; inter-individual variation limits generalisation; non-invasive accuracy remains below FDA-recognised clinical standards.	PPG- and ECG-based sensing combined with deep learning shows real promise for glucose *trend* detection, but not yet for clinical-grade absolute measurement. Integration with mobile platforms is a viable direction.	Narrative and systematic literature review; technology benchmarking; ML method comparison against FDA evaluation metrics.	The key source for this thesis's position on mobile sensing: it supports smartphone PPG as a plausible trend signal while documenting that non-invasive glucose accuracy remains below clinical standards. That distinction is exactly why the design retains a glucometer for glucose and treats mobile sensors as exploratory. *→ Chapter 2 §2.4 (IoT/Sensor Technology); Chapter 3 §3.4; Limitations (Technological)*
12	**Liu et al. (2025) — Biomedical Engineering Online** "A Finger on the Pulse of Cardiovascular Health: Estimating Blood Pressure with Smartphone Photoplethysmography-Based Pulse Waveform Analysis" Liu, I. S., Liu, F., Zhong, Q., & Ni, S. (2025). A finger on the pulse of cardiovascular health: Estimating blood pressure with smartphone photoplethysmography-based pulse waveform analysis. *Biomedical Engineering Online, 24*(1), 36. https://doi.org/10.1186/s12938-025-01365-w	Smartphone-based photoplethysmography blood pressure estimation remains limited by weak accuracy and by machine learning predictions that are not interpretable to clinicians.	A smartphone-only (finger-camera) PPG framework for cuffless blood pressure estimation, combining waveform feature engineering with SHAP analysis so predictions trace back to interpretable physiological features.	Improve the accuracy of smartphone-derived PPG blood pressure estimation while keeping the underlying model interpretable and clinically explainable.	Modest sample (127 participants); single-site data collection; smartphone PPG accuracy still trails clinical-grade cuff devices; generalisation across skin tones, devices and lighting not fully established.	Smartphone-derived PPG waveform features correlated strongly with reference blood pressure devices, and SHAP analysis gave transparent, feature-level explanations for predictions, supporting smartphone PPG as a feasible low-cost proxy for continuous BP monitoring.	Feature engineering from smartphone camera PPG waveforms; ML regression/classification; SHAP-based interpretability; Bland–Altman agreement analysis against reference devices.	The closest technical precedent for the "mobile sensors" data source named in this thesis's title: a smartphone camera alone, with no additional hardware, produces clinically meaningful blood-pressure signal. Its use of SHAP to interpret waveform-level predictions also reinforces the choice to report feature importance as evidence for H2. *→ Chapter 2 §2.4 (IoT/Sensor Technology); Chapter 3 §3.4 (Data Fusion); Chapter 5 (Feature Importance)*

CLUSTER 4 — MACHINE LEARNING FOR CLINICAL RISK PREDICTION

#	Source & Citation	Problem Addressed	Research Contribution	Aim & Objectives	Limitations	Findings & Conclusions	Methods	My Abstract / Relevance
13	**Mendoza-Cano et al. (2025) — PLOS ONE** "Data-Driven Identification of Key Predictors of Uncontrolled Hypertension: A Cross-Sectional Study" Mendoza-Cano, O., Trujillo, X., Huerta, M., Ríos-Silva, M., Lugo-Radillo, A., Bricio-Barrios, J. A., Benites-Godínez, V., Cuevas-Arellano, H. B., Uribe-Ramos, J. M., Solano-Barajas, R., Venegas-Ramírez, J., Ríos-Bracamontes, E. F., García-Solórzano, L. A., Camacho-delaCruz, A. A., & Murillo-Zamora, E. (2025). Data-driven identification of key predictors of uncontrolled hypertension: A cross-sectional study. *PLOS ONE*. https://doi.org/10.1371/journal.pone.0331565	Traditional statistical approaches under-identify which clinical and demographic factors actually drive poor blood pressure control, limiting the ability to target interventions.	A comparative analysis applying LASSO regression, Random Forest and XGBoost with SHAP interpretability to identify and rank predictors of uncontrolled hypertension, benchmarked against traditional logistic regression.	Identify key predictors of uncontrolled hypertension using data-driven ML methods; compare predictive performance of ensemble models against traditional logistic regression.	Cross-sectional design precludes causal inference; single-country cohort (Mexico); self-reported diagnosis history; possible residual confounding not captured in survey variables.	Random Forest outperformed both XGBoost and logistic regression, with years since diagnosis and age as the strongest predictors; SHAP-based interpretation corroborated age as a leading driver. Ensemble ML with interpretability tools can outperform classical statistical models for hypertension risk stratification.	Cross-sectional analysis of national health survey data; LASSO feature selection; Random Forest and XGBoost classification; SHAP value analysis; AUC-based model comparison.	A close methodological precedent for this thesis's core contribution: it validates Random Forest and XGBoost, benchmarked against a classical baseline, for hypertension risk classification, and confirms that interpretability tooling is expected practice in this space. Its finding that ensemble tree models outperform logistic regression supports the model choice and reinforces H1. *→ Chapter 3 §3.5–3.6 (Model Architecture / Feature Importance); Chapter 5 (H1/H2 Validation)*
14	**Efendi et al. (2025) — Health Science Reports** "IoT-Based Elderly Health Monitoring System Using Firebase Cloud Computing" Efendi, A., Ammarullah, M. I., Isa, I. G. T., Sari, M. P., Izza, J. N., Nugroho, Y. S., Nasrullah, H., & Alfian, D. (2025). IoT-based elderly health monitoring system using Firebase cloud computing. *Health Science Reports, 8*(3), e70498. https://doi.org/10.1002/hsr2.70498	Continuous health monitoring for elderly patients lacks validated IoT implementations using cloud platforms such as Firebase for real-time status classification.	Designs and validates an IoT and Firebase elderly monitoring system with a mobile interface and supervised machine learning classifying the user as "stable" or "not stable" from real-time parameters.	Develop and validate an IoT-Firebase health monitoring system; determine the optimal supervised ML model for deployment; evaluate device accuracy and user satisfaction.	Small validation sample (six participants); controlled conditions; monitors heart rate, oxygen saturation and body temperature — not blood pressure or glucose, so the parameter set differs from this thesis.	The system achieved a mean absolute percentage error of 0.90% across parameters (heart rate 1.68%, SpO2 0.57%, temperature 0.44%). In the comparative machine learning experiment, **XGBoost was the optimal model**. Firebase Firestore was validated as a real-time backend for clinical monitoring.	Three-layer IoT architecture (physical, network, application); device validation via MAPE against reference instruments; comparative supervised ML benchmarking; user satisfaction evaluation.	The closest technical precedent to this system's architecture. It validates Firebase Firestore as a real-time clinical backend and independently selects XGBoost as the optimal classifier — the same model chosen here, arrived at separately. State that its parameter set differs so the precedent is not overclaimed. *→ Chapter 2 §2.4 (IoT Architecture); Chapter 3 (Model Choice); Chapter 4 §4.4 (Technology Stack)*

CLUSTER 5 — DIGITAL HEALTH AND TELEMEDICINE IN SUB-SAHARAN AFRICA

#	Source & Citation	Problem Addressed	Research Contribution	Aim & Objectives	Limitations	Findings & Conclusions	Methods	My Abstract / Relevance
15	**Bediang et al. (2021) — Studies in Health Technology and Informatics** "Evaluation of the Effectiveness of Telemedicine in the Management of Cardiovascular Diseases in Primary Health Care in Cameroon: An Interventional Study" Bediang, G., Nganou-Gnindjio, C. N., Kamga, Y., Goethe Doualla, F.-C., Bagayoko, C. O., & Nko'o, S. (2021). Evaluation of the effectiveness of telemedicine in the management of cardiovascular diseases in primary health care in Cameroon: An interventional study. *Studies in Health Technology and Informatics, 281*, 615–619. https://doi.org/10.3233/SHTI210244	Lack of controlled, in-country evidence on whether telemedicine actually improves cardiovascular and hypertension care delivered at Cameroonian primary health facilities, as opposed to evidence imported from higher-resource settings.	A controlled multicenter interventional study comparing two Cameroonian primary care facilities with tele-ECG and tele-expertise support against two control facilities without it.	Evaluate whether remote clinician tele-expertise support to primary care providers improves clinical processes, clinical outcomes and patient satisfaction for cardiovascular disease management in Cameroon.	Only four facilities in total; short intervention window; findings centre on the tele-ECG cardiovascular use case rather than combined hypertension-diabetes monitoring; limited statistical power.	Facilities with tele-expertise support showed improved clinical processes, better patient outcomes and higher patient satisfaction than control facilities, supporting the feasibility and effectiveness of telemedicine in Cameroonian primary care.	Controlled multicenter interventional study; intervention versus control primary health facilities in Cameroon.	The most directly relevant in-country evidence available: a controlled Cameroonian study showing that remote clinician support for chronic cardiovascular conditions works in the exact primary-care setting this thesis targets. It grounds the deployment rationale and the clinician-dashboard component in local evidence rather than general SSA literature alone. → *Chapter 1 §1.1–1.2 (Background/Problem Statement); Chapter 2 §2.2; Chapter 4 (Deployment Context)*
16	**Dodoo, Al-Samarraie & Alsswey (2022) — Health and Technology** "The Development of Telemedicine Programs in Sub-Saharan Africa: Progress and Associated Challenges" Dodoo, J. E., Al-Samarraie, H., & Alsswey, A. (2022). The development of telemedicine programs in Sub-Saharan Africa: Progress and associated challenges. *Health and Technology, 12*(1), 33–46. https://doi.org/10.1007/s12553-021-00626-7	Despite growing interest in telemedicine across Sub-Saharan Africa, systematic documentation of implementation progress and region-specific barriers is lacking.	Review of 66 empirical studies documenting telemedicine progress across SSA countries and classifying the barriers by type.	Document the state of telemedicine development in SSA; identify barriers and facilitators; propose policy and design recommendations.	Rapid change in the field means findings date quickly; uneven representation of Francophone African countries, including Cameroon.	Progress across SSA is markedly unbalanced. Technological, organisational, legal/regulatory, individual, financial and cultural factors were identified as the major barriers to telemedicine success in the region.	Systematic review of 66 empirical studies; thematic classification of barriers.	Essential context for the deployment environment. The barrier taxonomy — connectivity, digital literacy, infrastructure — directly shaped the offline-first design and simplified UX. The noted under-representation of Francophone Africa is itself an argument for a Cameroon-sited study. → *Chapter 1 §1.2 (Problem Statement); Chapter 2 §2.2.3; Chapter 4 (Design Rationale)*
17	**Segun-Omosehin et al. (2025) — npj Cardiovascular Health** "Evaluating the Digital Health Technology Landscape in Sub-Saharan Africa and Its Implications for Cardiovascular Health" Segun-Omosehin, O. A., Omiye, J. A., Elalfy, A., Moideen-Sheriff, S., Kuti, F., Omole, O., Ogunyankin, K. O., Idemili-Aronu, N., Olawepo, J. O., Ezeanolue, E. E., & Adedinsewo, D. (2025). Evaluating the digital health technology landscape in sub-Saharan Africa and its implications for cardiovascular health. *npj Cardiovascular Health, 2*, 20. https://doi.org/10.1038/s44325-025-00055-9	Digital health technologies are expanding rapidly in SSA, but their implications specifically for cardiovascular care had not been drawn together in one assessment.	Perspective mapping the digital health technology and AI landscape across SSA and identifying where these tools can realistically be applied in cardiovascular medicine, including barriers to adoption.	Survey digital health technology use across SSA; assess applicability to cardiovascular medicine; identify implementation gaps and opportunities.	Perspective rather than systematic review, so coverage is selective rather than exhaustive; a fast-moving landscape; regulatory environments differ widely across SSA countries.	SSA faces poor access to care, limited human resources and weak health information systems, but health technology is among the region's top digital investment sectors. IoT and mobile telephony are identified as the most promising modalities for healthcare impact.	Perspective and narrative landscape assessment with structured coverage of digital health technology categories and cardiovascular applications.	Current, geographically relevant evidence for the digital health gap this study addresses, and authoritative support for the mobile-first, IoT-oriented modality choice. Its emphasis on weak health information management systems aligns closely with the fragmented record-keeping measured in this study's own cohort. Cite for landscape context, not as pooled evidence. → *Chapter 1 §1.1 (Background); Chapter 2 §2.2 (SSA Burden); Chapter 7 (Significance)*
18	**Agbeyangi & Lukose (2025) — Healthcare (MDPI)** "Telemedicine Adoption and Prospects in Sub-Sahara Africa: A Systematic Review with a Focus on South Africa, Kenya, and Nigeria" Agbeyangi, A. O., & Lukose, J. M. (2025). Telemedicine adoption and prospects in Sub-Sahara Africa: A systematic review with a focus on South Africa, Kenya, and Nigeria. *Healthcare, 13*(7), 762. https://doi.org/10.3390/healthcare13070762	Telemedicine adoption rates, policy frameworks and implementation prospects in SSA are insufficiently documented in comparative, multi-country analyses.	Systematic review of 53 studies (from 567 screened, 2014–2024) comparing telemedicine adoption, implementation and technological prospects across South Africa, Kenya and Nigeria.	Document telemedicine adoption status across three leading SSA countries; identify common barriers; derive comparative, actionable policy and design insights.	Three-country focus may not represent Cameroon or Central Africa; rapid policy evolution may limit currency; reliance on published literature only.	South Africa leads adoption, including specialist teleconsultation and chronic disease management; Kenya shows strong mHealth integration; Nigeria is advancing remote diagnosis under infrastructural constraint. Persistent barriers are regulatory gaps, digital literacy and infrastructure.	PRISMA-guided systematic search of PubMed, Scopus and Web of Science; structured critical appraisal for bias; thematic analysis; bibliometric keyword co-occurrence mapping.	Regional context for deployment: chronic disease management is named explicitly as a leading telemedicine use case in the region's most advanced country, while the barrier set justifies the offline-first architecture and simplified UX. The absence of Central Africa from the comparison is a gap this thesis speaks into. → *Chapter 1 §1.2 (Problem Statement); Chapter 2 §2.2.3; Chapter 4 §4.8 (Ethics & Deployment)*

CLUSTER 6 — DATA PRIVACY, SECURITY AND ETHICS IN DIGITAL HEALTH

#	Source & Citation	Problem Addressed	Research Contribution	Aim & Objectives	Limitations	Findings & Conclusions	Methods	My Abstract / Relevance
19	**Alhammad et al. (2024) — Journal of Medical Internet Research** "Patients' Perspectives on the Data Confidentiality, Privacy, and Security of mHealth Apps: Systematic Review" Alhammad, N., Alajlani, M., Abd-alrazaq, A., Epiphaniou, G., & Arvanitis, T. (2024). Patients' perspectives on the data confidentiality, privacy, and security of mHealth apps: Systematic review. *Journal of Medical Internet Research, 26*, e50715. https://doi.org/10.2196/50715	No prior review had consolidated patients' own perceptions of, and concerns about, the confidentiality, privacy and security of personal health data collected by mHealth apps — a known factor in adoption.	A PRISMA-guided systematic review of 33 studies synthesising patient perspectives, awareness and behaviour regarding mHealth data privacy, confidentiality and security, including the internal and external risk factors shaping trust.	Investigate patients' perspectives and awareness of confidentiality, privacy and security of data collected through mHealth apps, and identify factors associated with these perceptions.	Included studies span 2014–2022 and are heterogeneous in method and population; most originate from high-income countries; relies on self-reported perception rather than objective breach or incident data.	Patient concerns about data confidentiality, privacy and security are a consistent, significant barrier to mHealth adoption. Encryption, secure authentication and regular risk assessment are recommended baseline safeguards, and transparency about data handling improves patient trust.	PRISMA-guided systematic review; multi-database search (PubMed, Ovid, ScienceDirect); thematic synthesis of patient-perspective studies.	Since this thesis processes real, hospital-sourced patient data through a mobile and cloud pipeline, this review supplies the evidence base for treating data confidentiality and security as a first-class design constraint. It supports the ethics and deployment discussion around anonymisation, secure storage and patient trust, and justifies the authentication and encrypted transmission choices in the technology stack. → *Chapter 1 §1.2 (Problem Statement — data governance); Chapter 4 §4.8 (Ethics & Deployment)*

Quick Reference Summary — All 19 Sources

#	Reference	Chapter(s) Where Cited	Thematic Cluster
1	Hinneh et al. (2023) — PLOS Glob Public Health 3(12) e0001931	Ch.1 §1.1; Ch.2 §2.1–2.2	Epidemiological Burden
2	Gafane-Matemane et al. (2024) — J Hum Hypertens 39(2) 95–110	Ch.1 §1.1; Ch.2 §2.2	Epidemiological Burden
3	World Health Organization (2023) — Global Report on Hypertension	Ch.1 §1.1; Ch.2 §2.1	Epidemiological Burden
4	Mihevc et al. (2025) — JMIR Mhealth Uhealth 13, e59733	Ch.2 §2.3; Ch.6	mHealth & Clinical Validation
5	Mao et al. (2020) — BMJ Open Diab Res Care 8(1) e001225	Ch.2 §2.3; Abstract	mHealth & Clinical Validation
6	Bandeira et al. (2024) — Telemed e-Health 30(4) 987–1001	Ch.2 §2.3; Ch.6 §6.6	mHealth & Clinical Validation
7	Alessa et al. (2019) — JMIR Mhealth Uhealth 7(5) e13645	Ch.2 §2.3; Ch.3	mHealth & Clinical Validation
8	Zhou et al. (2024) — J Med Internet Res 26, e54978	Ch.2 §2.3; Ch.3; Ch.5	mHealth & Clinical Validation
9	Tai et al. (2024) — Trans Emerging Tel Tech 35(4) e4919	Ch.2 §2.4; Ch.3 §3.5	IoT / Sensor Technology
10	Liu & Wang (2025) — Front Public Health 13, 1510456	Ch.2 §2.4–2.5; Ch.3 §3.5	IoT / Sensor Technology
11	Alhaddad et al. (2022) — Front Bioeng Biotechnol 10, 876672	Ch.2 §2.4; Ch.3 §3.4	IoT / Sensor Technology
12	Liu et al. (2025) — Biomed Eng Online 24(1) 36	Ch.2 §2.4; Ch.3 §3.4; Ch.5	IoT / Sensor Technology
13	Mendoza-Cano et al. (2025) — PLOS ONE	Ch.3 §3.5–3.6; Ch.5	ML for Risk Prediction
14	Efendi et al. (2025) — Health Sci Rep 8(3) e70498	Ch.2 §2.4; Ch.3; Ch.4 §4.4	ML for Risk Prediction
15	Bediang et al. (2021) — Stud Health Technol Inform 281, 615–619	Ch.1 §1.1–1.2; Ch.2 §2.2; Ch.4	Digital Health SSA (Cameroon)
16	Dodoo, Al-Samarraie & Alsswey (2022) — Health Technol 12(1) 33–46	Ch.1 §1.2; Ch.2 §2.2.3; Ch.4	Digital Health SSA
17	Segun-Omosehin et al. (2025) — npj Cardiovasc Health 2, 20	Ch.1 §1.1; Ch.2 §2.2; Ch.7	Digital Health SSA
18	Agbeyangi & Lukose (2025) — Healthcare 13(7) 762	Ch.1 §1.2; Ch.2 §2.2.3; Ch.4 §4.8	Digital Health SSA
19	Alhammad et al. (2024) — J Med Internet Res 26, e50715	Ch.1 §1.2; Ch.4 §4.8	Data Privacy & Ethics

Nineteen sources spanning epidemiological burden, mHealth self-management and clinical outcome validation, IoT and mobile sensor technology, machine learning for clinical risk prediction, digital health and telemedicine in Sub-Saharan Africa including Cameroon-specific evidence, and data privacy and ethics — covering all thematic areas required by Chapter 2, together with the dataset, contribution and limitations questions raised in supervision.

One further source is used in the methodology but is not listed here because it is a clinical guideline rather than a research article: the **2023 ESH Guidelines for the Management of Arterial Hypertension** (Mancia, G., Kreutz, R., Brunström, M., et al., *Journal of Hypertension, 41*(12), 1874–2071, doi 10.1097/HJH.0000000000003480), from which the blood pressure thresholds used to define the risk levels are taken. Add it to the bibliography as a guideline, and cite it wherever the thresholds are stated.